

PHC Integrated Digital System

Groupno_Lab910

Lab 9 & 10 Report: Software Metrics and Risk Handling

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Report intent. This report tells the project story as one connected narrative: how the PHC system evolved sprint by sprint, what the repository proves today, which metrics matter, where the real risks emerged, and how those risks were already folded back into planning and backlog decisions.

1. Project Story and Evidence Baseline

The PHC Integrated Digital System started as a documentation-heavy academic project, but by the end of Sprint 5 it had become a working backend platform with authentication, patient records, visit lifecycle management, prescription handling, laboratory flows, billing, appointments, notifications, and admin reporting. That progression matters for this lab because the requested metrics are meaningful only if they are tied to the actual engineering story rather than to abstract estimates.

For this report, we used the current repository as the evidence baseline. The measured story comes from `README.md`, `TESTING.md`, `test_cases.csv`, `backend/prisma/schema.prisma`, all backend route and service files, SRS revisions through Version 2.2, and git history from January 15, 2026 to April 3, 2026. The resulting snapshot is:

Repository-backed baseline	Measured value
Production source lines of code	2859
Backend source lines of code	2505
Service-layer source lines of code	1608
Persisted domain models	20
Implemented API endpoints	57
Documented categorized test cases	48
Feature commits in the active project window	21

The most useful qualitative observation from this baseline is that the project is backend-heavy by design. The service layer contains most of the domain logic, which is exactly what we would expect from a system where correctness, traceability, and role safety matter more than early frontend polish.

1.1 Late Sprint 5 Closeout Evidence

Because Sprint 5 was the backend integration sprint, the latest repository evidence is concentrated near the end of March 2026. The sequence below is the corrected chronology used for the tracker and for this report.

Date	Commit	Sprint 5 item	Meaning in the project story
2026-03-25	9e3e7c0	Local LDAP-backed authentication	Converted authentication from the earlier dev fallback approach into a proper local LDAP bind flow for seeded users.
2026-03-25	3fac9c2	Full-system E2E runner	Added a repository-backed end-to-end validation path across the major back-end modules.
2026-03-25	2d7e421	Sprint 5 integration docs closeout	Synchronized API documentation and backend integration evidence near sprint closeout.
2026-03-25	9182fcc	Availability notifications	Added doctor unavailability and in-app notification logic, reducing operational ambiguity.
2026-03-24	5ff16c2	Admin user management	Added administrative account creation, listing, status control, and role updates.
2026-03-24	248d70a	Usage and attendance reports	Added management-facing reporting endpoints for usage and doctor attendance summaries.
2026-03-24	b66f7f2	PHC events module	Added event publishing and public event visibility for PHC communication.
2026-03-24	7718df7	Patient lab report access	Added role-safe patient access to their own laboratory reports.
2026-03-24	f1ef67b	Lab request detail endpoint	Added single-request detail retrieval with access control checks.

2. Section I: Metrics and Their Meaning

2.1 Intermediate COCOMO

We modeled the project as an *Organic* COCOMO system because the team is small, the stack is familiar, and the product is complex enough to require engineering discipline without becoming an embedded or mission-critical real-time system. Using 2.859 KLOC of production code and an effort adjustment factor derived from reliability, data sensitivity, functional complexity, team capability, and tooling assumptions, we obtained the following result:

COCOMO measure	Value
Mode	Organic
KLOC	2.859
Effort Adjustment Factor (EAF)	0.786
Estimated effort	7.58 person-months
Estimated schedule	5.40 months
Average staffing	1.40 persons
Productivity	0.377 KLOC / PM

This estimate aligns with the real project story better than it may first appear. The backend has been completed by a small team over a compressed academic schedule, but that happened by deferring

frontend work and concentrating effort in the backend first. So the main use of COCOMO here is not to claim exact planning accuracy; it is to show that the system has already crossed the scale of a toy application and sits in the range where architectural discipline matters.

2.2 Halstead Metric

Halstead metrics were computed over the backend source files after removing comments and string literals. The purpose was to estimate the size and mental complexity of the implemented codebase.

Halstead measure	Value
Distinct operators n_1	55
Distinct operands n_2	536
Total operators N_1	11799
Total operands N_2	4931
Program vocabulary	591
Program length	16730
Volume	154033.35
Difficulty	252.99
Effort	38968856.88
Estimated delivered bugs	51.34

The most important takeaway is not the absolute effort number but what it implies: the backend already has enough vocabulary and domain spread that ad hoc coding would quickly become fragile. That explains why the current layered structure, route-level RBAC, and transaction-protected operations are not optional niceties; they are required to keep the system understandable.

2.3 Function Point Analysis

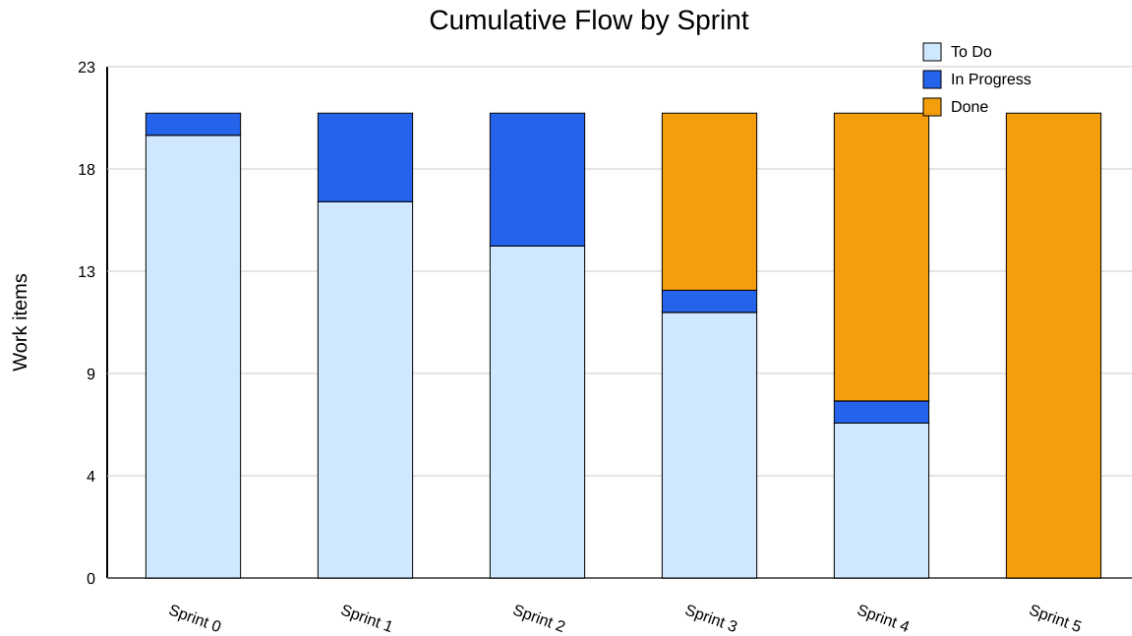
Function Point Analysis was derived from the actual route inventory and database model. Endpoints were classified into external inputs, outputs, and inquiries; Prisma models were treated as internal logical files; LDAP was treated as one external interface file.

Type	Count	Weight	Contribution
EI	28	4	112
EO	4	5	20
EQ	25	4	100
ILF	20	10	200
EIF	1	5	5
UFP	437		
VAF	1.11		
Adjusted FP	485.07		

This metric reinforces a pattern already visible in the codebase: the system is not just doing CRUD on a few entities. It has workflow-bearing transactions, state transitions, and role-gated behavior. That is exactly why the project narrative should be told as a connected system story and not as disconnected endpoint counts.

2.4 Cumulative Flow Diagram

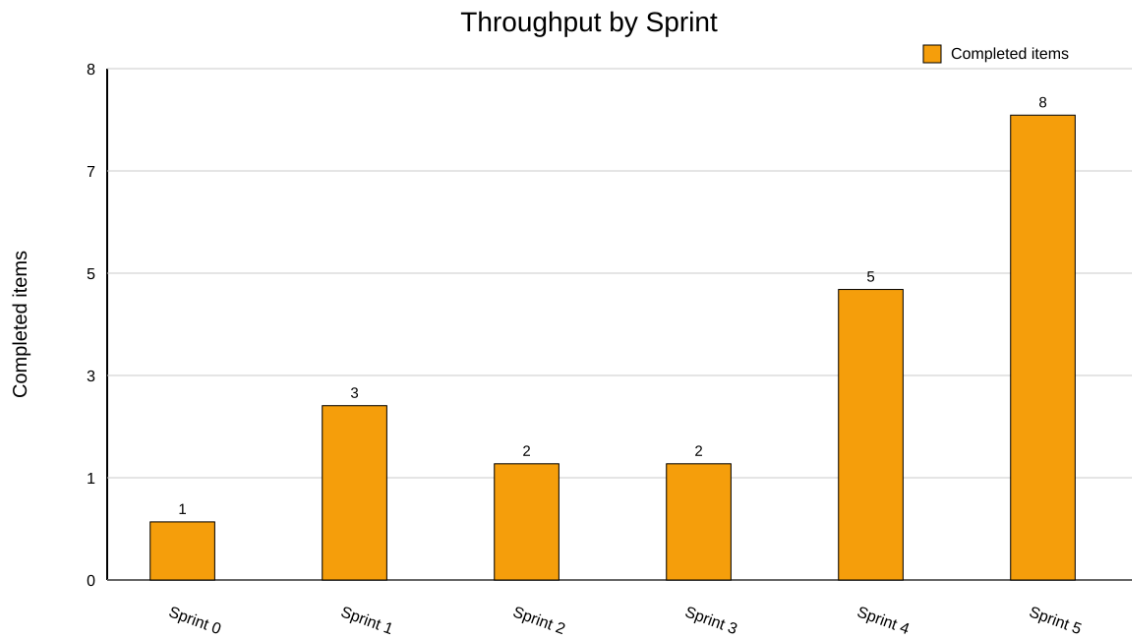
The repository does not store Trello history directly, so the cumulative flow view was reconstructed from the sprint plan in the SRS/README and the actual implementation completion pattern visible in git. The figure below summarizes how planned work transitioned from *To Do* to *In Progress* to *Done* across Sprint 0 to Sprint 5.



The connected story here is that work did not close uniformly. The CFD shows that a significant amount of scope remained in progress until later sprints, which means the team was relying on staged maturity: foundation first, workflow next, operational modules after that, and only then full integration. That makes sense for this project, but it also explains why Sprint 5 felt dense.

2.5 Throughput Report

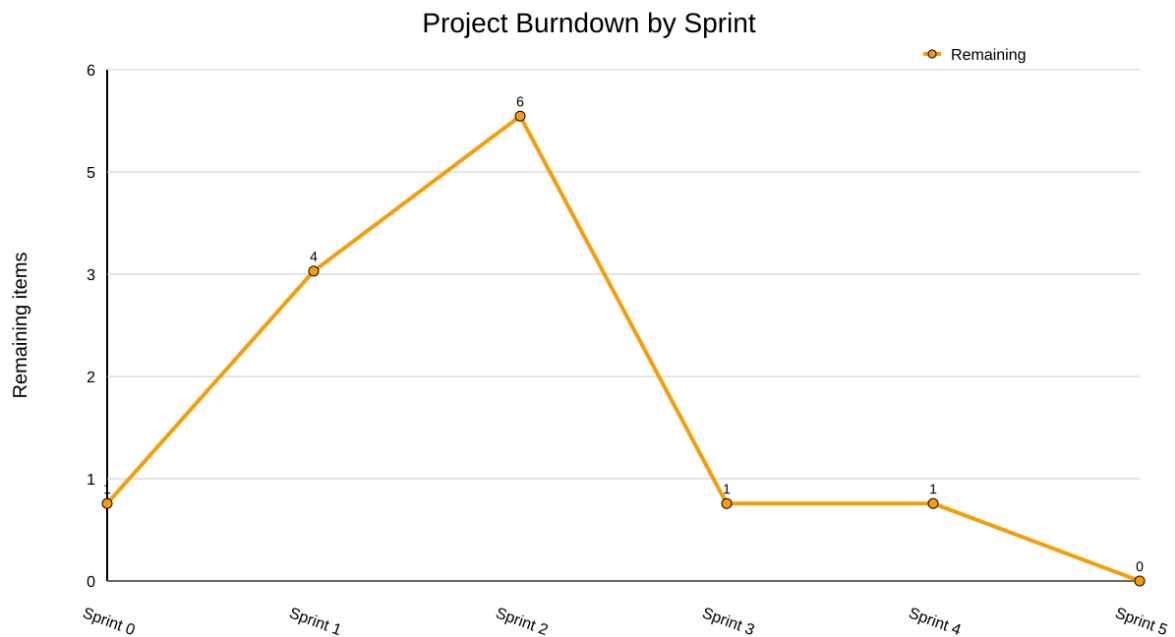
Throughput was measured at two levels: completed work items by sprint and feature-delivery behavior by git history. The project-wide sprint throughput shape is shown below.

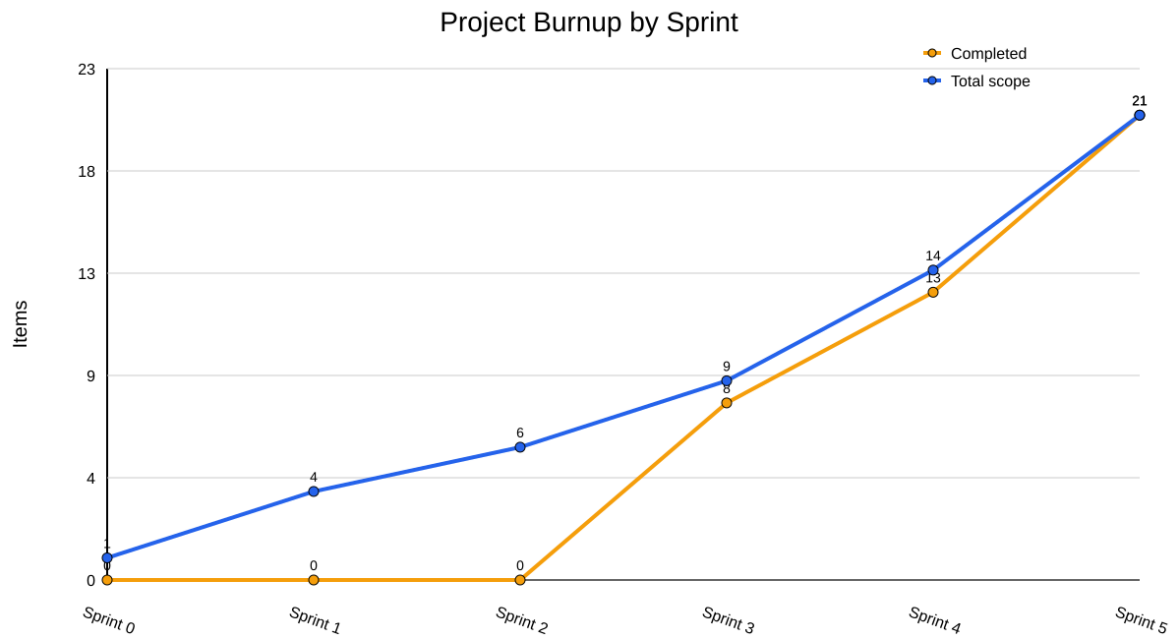


The important observation is that throughput increases toward Sprint 4 and Sprint 5 instead of being flat across the semester. That is a real signal, not noise. It tells us that once the backend architecture stabilized, the team accelerated feature closure. The cost of that acceleration, however, was a more compressed integration phase.

2.6 Project Burndown and Burnup

Because the prompt explicitly allows telling the test and project story in our own way, we chose project-wide burn charts instead of a narrow single-sprint chart. This gives a more honest picture of how scope accumulated and how completion caught up over time.

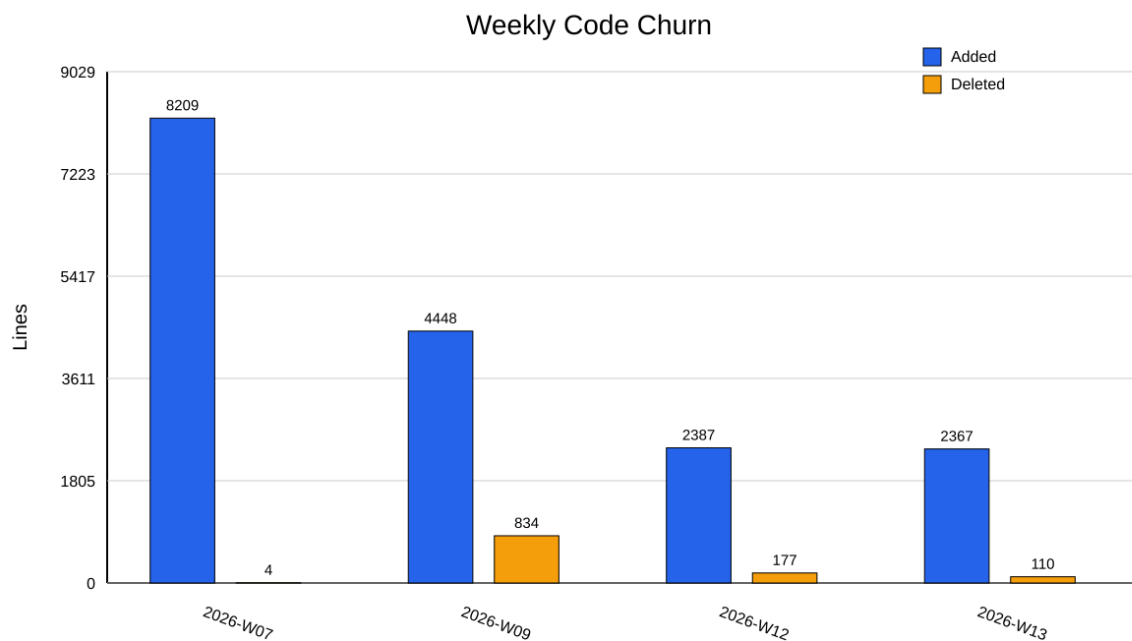




The burndown and burnup together tell a coherent story. Scope expanded in a planned way from Sprint 0 through Sprint 5, but completion rose in steps rather than smoothly. That means the main process weakness was not uncontrolled scope growth; it was late visible closure. This distinction matters because the mitigation is not to reduce ambition, but to make integration and documentation visible earlier.

2.7 Additional Metric 1: Code Churn

Code churn was measured from git additions and deletions aggregated by week.

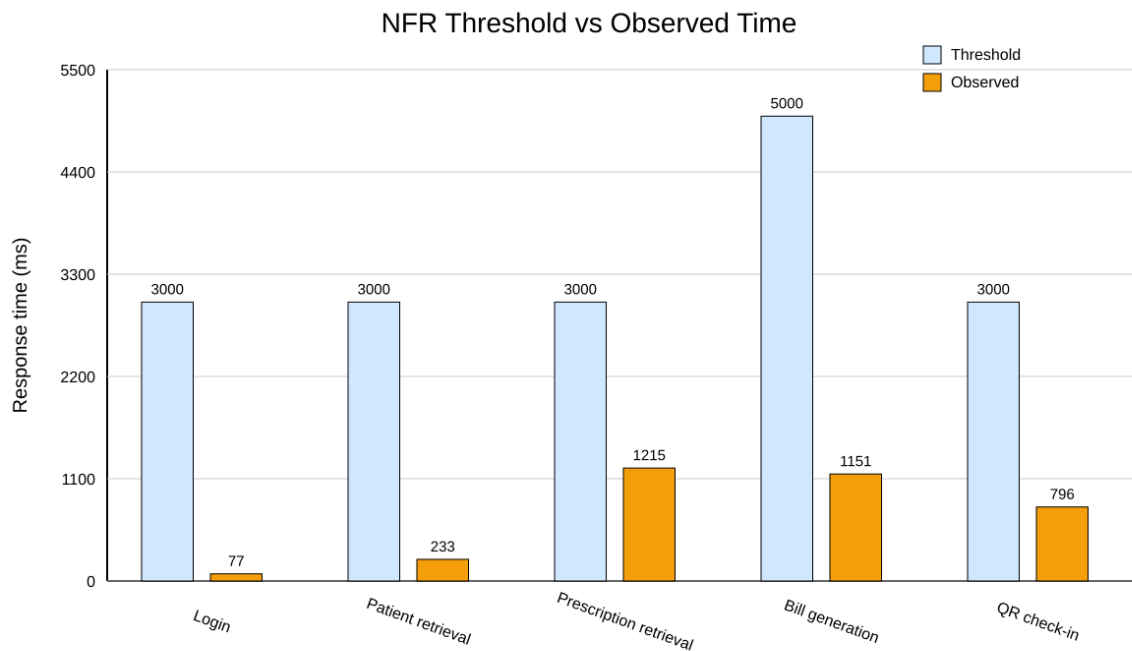


The project had its highest churn during the phase where the backend moved from setup to real implementation. Later churn remained substantial during operational-module work and Sprint 5

integration. This is healthy as long as it is paired with tests and documentation. Without that pairing, churn would instead be a sign of instability.

2.8 Additional Metric 2: Test and Performance Story

The second additional metric is a combined quality story: pass rate and NFR timing compliance. The repository currently documents 48 categorized test cases in the CSV, all marked PASS, while the broader smoke automation has grown to 59 checks.



The chart shows that all observed timings remain comfortably inside the stated thresholds. Login, patient retrieval, prescription retrieval, bill generation, and QR check-in all stay within the bounds defined in the project's testing narrative. The positive result is that functional maturity and performance sanity were achieved together. The caution is that documentation synchronization became a project risk once the smoke suite grew beyond the 48-case CSV.

3. Section II: Risk Analysis

3.1 Why a 5x5 Matrix Fits This Project

The attached template suggests a classic 5x5 probability-impact matrix. That is appropriate here because the PHC system is not dominated by one catastrophic technical unknown; instead, it faces multiple moderate-to-high operational and integration risks that need prioritization. We therefore use the following scale. **Probability:** 1 = Rare, 2 = Unlikely, 3 = Moderate, 4 = Likely, 5 = Almost Certain. **Impact:** 1 = Insignificant, 2 = Minor, 3 = Significant, 4 = Major, 5 = Severe. **Risk score:** $Probability \times Impact$.

3.2 5x5 Risk Matrix

5x5 Risk Matrix for PHC System					
	Impact <i>How severe would the outcome be if the risk occurred?</i>				
	Insignificant 1	Minor 2	Significant 3	Major 4	Severe 5
5 Almost Certain	Medium 5	High 10	Very high 15	Extreme 20	Extreme 25
4 Likely	Medium 4	Medium 8	High 12	Very high 16	Extreme 20
3 Moderate	Low 3	Medium 6	Medium 9	High 12	Very high 15
2 Unlikely	Very low 2	Low 4	Medium 6	Medium 8	High 10
1 Rare	Very low 1	Very low 2	Low 3	Medium 4	Medium 5

3.3 Project-Specific Risk Register

The matrix above is only the template. The real value comes from placing our own project risks into it.

ID	Risk	P	I	Score	Mitigation logic
R1	Cross-module integration regression across visits, lab, pharmacy, billing, and notifications	4	4	16	Keep transaction boundaries, maintain the E2E runner, and use smoke tests before documentation freeze.
R2	Late sprint convergence hides unfinished work until very close to milestone deadlines	4	4	16	Break work into smaller visible backlog items and insist on earlier integration checkpoints.
R3	LDAP or authentication instability blocks role-based access and demos	3	5	15	Use Dockerized local LDAP, explicit 401/503 behavior, seeded test users, and auth verification in smoke and E2E tests.
R4	Database migration or shared-model change damages visit-linked consistency	3	5	15	Use Prisma migrations only, preserve idempotent seed flow, and validate migrate deploy.

R5	Documentation drift between README, TESTING, smoke counts, CSV evidence, and report content	4	3	12	Treat docs as first-class deliverables and regenerate report data from scripts rather than manual counting.
R6	Frontend lag leaves the project backend-complete but demo-incomplete	4	3	12	Freeze backend scope unless blocked and turn the stable API into role dashboards in Sprint 6.
R7	RBAC error exposes medical data to the wrong role	2	5	10	Preserve route guards, add more RBAC tests, and keep patient-only endpoints clearly separated.
R8	Performance and security hardening remain behind functional completion	2	4	8	Reserve Sprint 7 for hardening, load checks, and deployment preparation rather than new features.

The probability assumptions here are not arbitrary. They are tied to what the repository already showed. For example, documentation drift is marked likely because the project already developed a mismatch between the 48-case CSV and the larger smoke suite. Late sprint convergence is marked likely because the burndown/burnup story shows completion clustering near Sprint 5 closeout.

4. Section III: Mapping Risk Mitigation to Sprint Planning

This final section is where the report becomes a connected story instead of a list of charts. The project did not merely *have* risks; it already responded to them through sprint ordering.

Sprint	Main risks addressed	How mitigation entered the backlog
Sprint 0	Scope ambiguity	Initial SRS revisions and UML work narrowed the project away from ambulance-style scope and toward a PHC-centered system.
Sprint 1	Auth, storage, and requirement instability	Authentication, schema, and patient profile work were pulled first so all later modules had a stable identity and persistence base.
Sprint 2	Workflow inconsistency	The team implemented visit lifecycle and doctor availability before prescriptions or labs, reducing downstream rework risk.
Sprint 3	Clinical-module fragmentation	Prescriptions and lab requests were added only after consultation and queueing existed, which kept clinical flows coherent.
Sprint 4	Transaction and operational risk	Billing and check-in were implemented with transaction-safe design, and operational modules were integrated without changing the visit-centric backbone.
Sprint 5	Integration, documentation, auth hardening	Admin modules, notifications, E2E flow validation, API documentation, and real local LDAP support were explicitly added as mitigation work, not just as features.
Sprint 6	Backend/UI imbalance	The current plan is to convert the stable API into role dashboards rather than reopen backend scope.
Sprint 7	Release-readiness risk	Testing depth, RBAC verification, security hardening, load checks, and deployment preparation are reserved for the stabilization sprint.

This is the clearest way to tell the project story: the sprint order itself is a risk-mitigation strategy. The team did not just build modules in arbitrary order. It used ordering to reduce structural risk first, workflow risk next, operational risk later, and integration risk near backend closeout.

5. Conclusion

The PHC Integrated Digital System now has a strong backend story backed by measurable evidence. The COCOMO estimate places it in a meaningful engineering range, the Halstead and function-point results show non-trivial complexity, the flow and throughput charts show how work matured over time, and the test/performance data shows that the backend is not only large enough to matter but also stable enough to demonstrate.

The deeper conclusion, however, is about project control. The team's real challenge is no longer *whether* it can build the backend. It already has. The real challenge is how to convert that backend maturity into a coherent final product by keeping documentation aligned, making Sprint 6 visible, and using Sprint 7 for hardening instead of feature creep. That is why this report tells one connected story: metrics, risks, and sprint planning are not separate topics in this project. They are three views of the same development reality.

Progress Tracker

Project tracker spreadsheet

AI Use Disclosure

AI assistance was used for report drafting, chart generation scripts, and workbook/report consolidation. All computed values, repository references, risk judgments, and final narrative were manually reviewed against the current project artifacts.